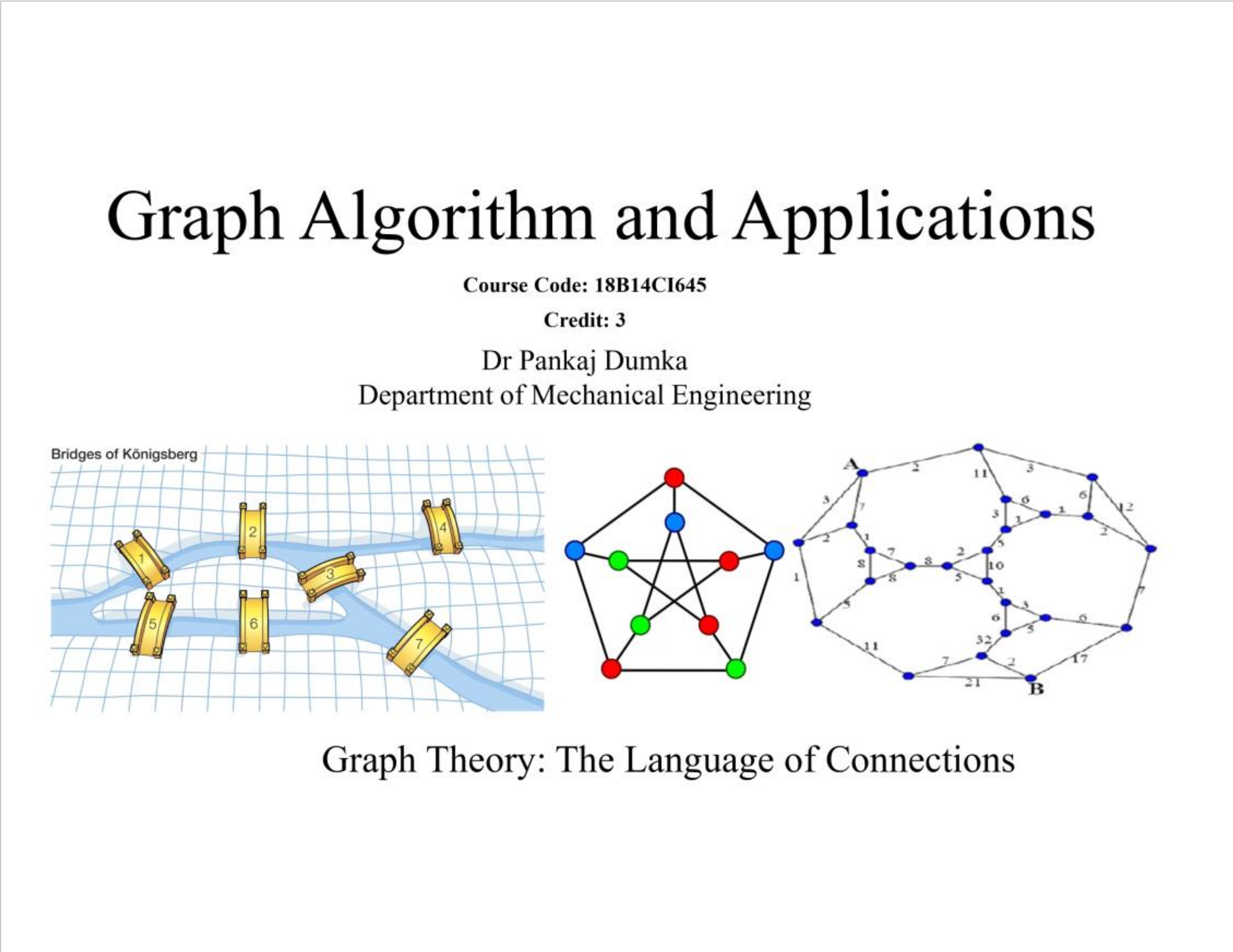
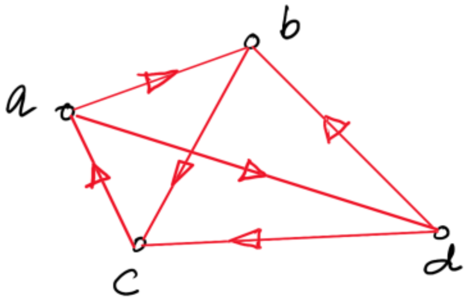
Graph Theory-L16-Tournament, Vertex colouring

01 October 2025 18:05



Digraphs

What we call this Graph?



Directed graph

OR

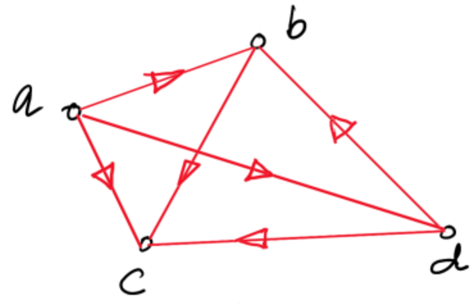
Digraph

They are represented by $\mathcal{D}$

Degree

d^+ | d^- | d_{Tot}

In digraphs the edges are called as **Arcs**.

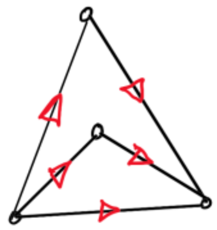


$\underline{\mathcal{D}}$

	d^+	d^-	d_{Tot}
a	0	3	3
b	2	1	3
c	3	0	3
d	1	2	3

Vertex for which indegree is zero is called as – source

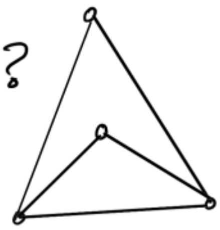
Vertex for which outdegree is zero is called as – sink



D

What is the difference b/w D & G?

Arcs

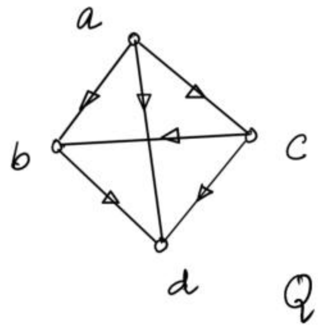


G

A graph which is obtained by removing directions from a digraph is called as Underlane graph.

Tournaments

*A **tournament** is a diagraph where every 2 vertices are joined by exactly one arc.*



Q:- underlane graph of a tournament is?

complete graph.

Q:- why the tournaments are imp?

Ans:- They are imp. to find the possible outcomes of a Round Robin match (tornnament).

Round-Robin Fixture

STANDINGS

	W	L
Archer 1	0	0
Archer 2	0	0
Archer 3	0	0

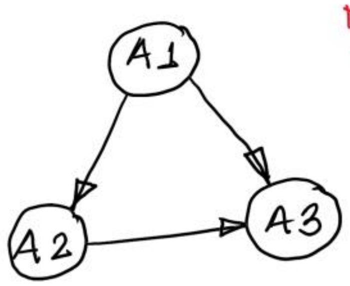
Every 2 players in the tournament must Play eachother.

SCHEDULE

Archer 2
Archer 3
Game 1 Edit Score Note

Archer 3
Archer 1
Game 2 Edit Score Note

Archer 1
Archer 2
Game 3 Edit Score Note



Note:- In round-robin tournament, no mach can be tied.

Q:- How winner's are represented?

winner is the one from which the arc originates.

$A_1 - A_2 \xrightarrow{\text{Winner}} A_1$

$A_1 - A_3 \xrightarrow{\text{Winner}} A_1$

$A_2 - A_3 \xrightarrow{\text{Winner}} A_2$

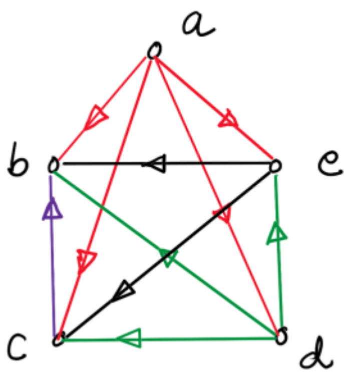
A_1	A_2	A_3
2	1	0

A_1 is the top player.

Observation? Score of a player = out degree of that vertex.

Five single tennis players ***a, b, c, d****, and* ***e****, were involved in a Round-Robin tournament. It was reported that:*

- ***a*** *defeated all other players*
- ***d*** *was defeated only by* ***a***
- ***b*** *was defeated by* ***c***
- ***e*** *defeated only 2 players.*
- *Are there any two players having the same score?*



a	b	c	d	e
4	0	1	3	2

Players having same score :-

No 2 players have same score.

Note:- In any tournament the total win should be:-

$$\binom{n}{2} \rightarrow \frac{n(n-1)}{2} = \sum \bar{d}(v_i)$$

In a 7-player tournament the recorded out-degrees (wins) of six players are 5,4,3,3,2,1. Find the out-degree of the 7th player.

$$\text{Total wins} = \binom{n}{2} = \frac{n(n-1)}{2}$$

$$n=7 \;:\; T_{win} = \frac{7\times 6}{2} = 21$$

$$5+4+3+3+2+1+x = 21$$

$$\boxed{x=3}$$

Determine whether the following nonincreasing sequence is a valid tournament score sequence for n=6:

$$[5,4,3,2,1,0]$$

$n=6$

Total wins $\rightarrow \binom{n}{2} = \frac{6\times 5}{2} = 15$

Σ Wins $= 5+4+3+2+1+0$

$= 15$

Hence, Possible

Graph Coloring

Vertex Coloring

Assigning colors to the vertices of a particular graph.

Depending on the no. of colours that I assigned to the vertices : the coloring is called as vertex K-coloring.

Colors used in graph	K-coloring
● ● ● $\longrightarrow$	3-coloring
● ● $\longrightarrow$	2-coloring
● ● ● ● $\longrightarrow$	4-coloring

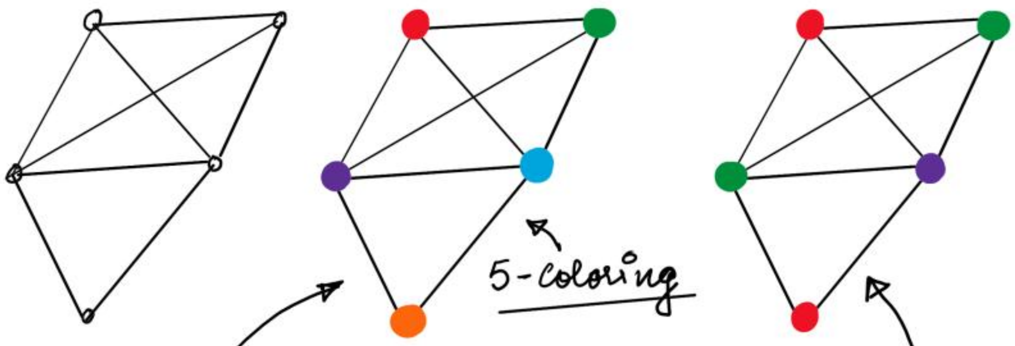
Coloring rule :- NO 2 adjacent vertices can have same color.

OR

Vertices should be colored such that their adjacent vertices have different colors.

This type of coloring is called as :- Proper vertex coloring

A ***proper vertex coloring*** *of a graph is a vertex coloring such that the end points of each edge are assigned 2 different colors.*



Q: Is this correct way of coloring?

A: Yes

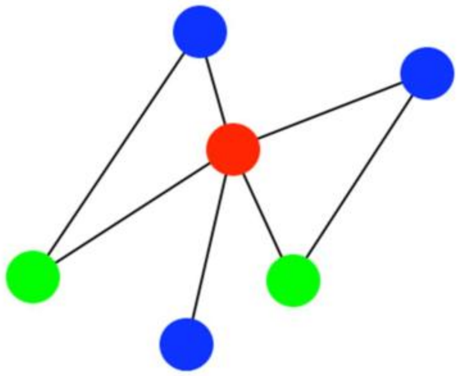
Q: why?

Q2: Is this correct way of coloring?

A: NO

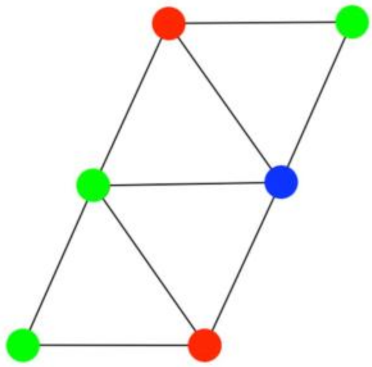
Q: why?

The Vertex Coloring below is a:



- 2-coloring
- 3-coloring
- 4-coloring

The graph given below is:

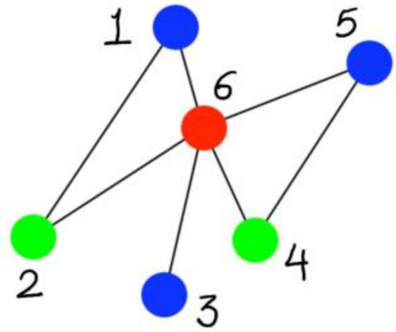


- 3-colorable
- 4-colorable
- Not properly colored

Only a ___________ graph can be vertex colored using one color.

- ○ **Complete**
- ○ **Null**
- ○ **Simple**

Color class :- Is a subset of the vertices of a graph that contains all the vertices of a given color.



$n = 6$

red $= \{6\}$

Blue $= \{1, 3, 5\}$

green $= \{2, 4\}$

Vertex $\underline{3}$ coloring

Chromatic Number

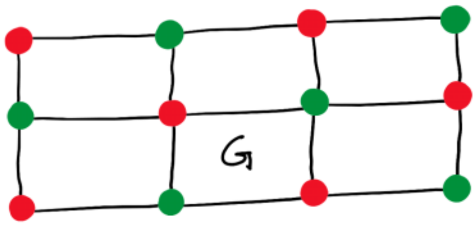
The chromatic no. of a graph G, denoted by $\chi(G)$, is the ***minimum no. of different colors*** *required for a proper vertex coloring of G.*

OR

The ***min. no. of colors*** *needed to paint all the vertices of a graph such that no two adjacent vertices have the same colour is called as the chromatic no. of G.*

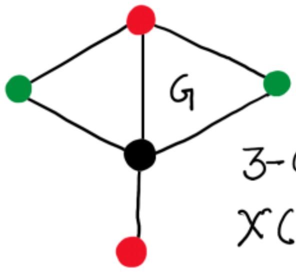
A graph G is (vertex) k-chromatic if $\chi(g) = k$.

Find the $\chi(G)$ for the following G's:



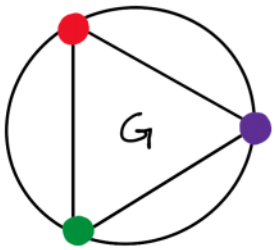
2 - chromatic

$\chi(G) = 2$



3-chromatic

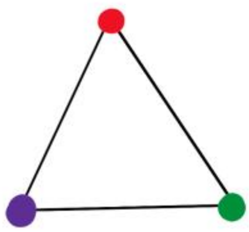
$\chi(G) = 3$



3-chromatic

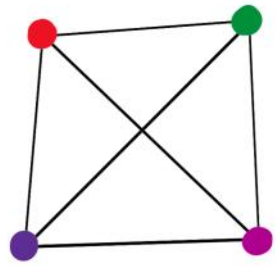
$\chi(G) = 3$

Find the $\chi(G)$ for the following G's:



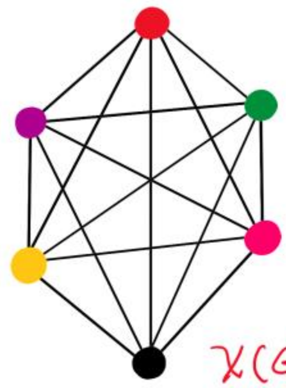
$\chi(G) = 3$

K_3



$\chi(G) = 4$

K_4



$\chi(G) = 6$

K_6

$K_n \longrightarrow n$-chromatic

observation:- A complete graph of n vertices is n-Chromatic

Q:- What is the chromatic no. for K_7 and K_{10}? 7 & 10

Color the vertices of this graph and tell the chromatic number (Homework Problem)